Anonymous Equal-Amplitude Composite-Wave Model: Basic Axiom System v6

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Author: Noriaki Kihara
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1. First Principles

Axiom 0: Component Anonymity

Classification: Axiom

Basic components are not assigned individual names, privileged axes, privileged signs, or privileged types.

The index `n` is a convenient label and is not an intrinsic name of the component.

Forbidden:

```
making only one specific component the negative-sign axis
making only one specific component the time axis
making only one specific component real-valued
making only one specific component complex-valued
giving an intrinsic name to only one specific component
```

Axiom 0+: Formulation Anonymity

Classification: Axiom

At the first-principle layer, theory names, formulation names, and readout-logic names are not assigned intrinsic names.

Equations, projections, or readout rules at the first-principle layer must not be changed on the basis of physical names, object names, interaction names, or existing theory classifications.

Forbidden:

```
adopting a circular-motion equation because it is called gravity
adopting a hyperbolic equation because it is called Coulomb force
erasing signs because it is called mass
preserving signs because it is called charge
moving a component to a time axis because it is called energy
using a different closure equation because the target is a two-body system
using a different closure equation because the target is a three-body system
using a special readout rule because the target is an ABC system
choosing a right-hand-side representation to obtain a desired physical name
```

Allowed:

```
different readout operations applied to the same closure condition
different symmetry breaking applied to the same closure condition
different gauge stability applied to the same closure condition
different sign preservation applied to the same closure condition
different sign erasure applied to the same closure condition
different phase-branch selection applied to the same closure condition
different closure target sets applied to the same closure condition
```

Order:

```
place the same all-positive-sign zero closure
define the readout operation first
define symmetry breaking first
define sign preservation or sign erasure first
confirm gauge stability
assign working names after readout
```

Target set:

$$Q(P) = \sum_n p_n^2$$

$$Q(A) = 0$$

$$Q(A + B) = 0$$

$$Q(A + B + C) = 0$$

Differences in target set are allowed.

Changing the definition of `Q` is not allowed.

Axiom 0.5: Dimension-Wise Central Projection Readout

Classification: Auxiliary geometric axiom

No curvature correction is introduced for readout along a single axis.

Curvature-correction candidates are introduced only when two or more independent directions form an area, volume, closed path, or transport mismatch.

Axiom 1: All-Positive-Sign Zero Closure

Classification: Axiom

$$\sum_{n=1}^{N} x_n^2 = 0$$

Expanded form:

$$x_1^2 + x_2^2 + \cdots + x_N^2 = 0$$

Coefficient signs:

```
all terms +
```

Not adopted:

$$\sum_n |x_n|^2 = 0$$

Term used for closure:

$$x_n^2$$

Term not used for closure:

$$x_n \bar{x}_n$$

Applicable states:

```
stationary closed state
stable closed state
```

Quasi-stationary states:

```
immediately after external influence
closure recovery process
metastable state
```

Temporary nonzero closure residual in a quasi-stationary state:

$$\sum_n x_n^2 \neq 0$$

After stationarization:

$$\sum_n x_n^2 = 0$$

Axiom 2: Nontrivial Existence

Classification: Axiom

$$\exists n, \quad x_n \neq 0$$

Axiom 3: Projection-Axis Degeneration and Imaginary-Direction Readout

Classification: Axiom

When an intrinsic coordinate system is constructed by a projection method, the projection-axis direction is not distinguishable as a direction for the residents of that intrinsic coordinate system.

However, if a value in the projection-axis direction remains inside the closure condition, that value is not erased.

When the compensating quantity in the projection-axis direction is read as `z`, it is treated inside the intrinsic coordinate system as

$$iz$$

This component is unobservable as a first-order direction.

When squared inside the closure expression,

$$(iz)^2 = -z^2$$

it appears with a reversed sign.

Therefore, an unobservable compensating quantity in the projection-axis direction appears inside the intrinsic coordinate system as a sign-reversed compensating quantity through squared readout.

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2. Consequences from the First Principles

Consequence 1: Insufficiency of Real Positive-Definite Components

Classification: Derived consequence

$$x_n \in \mathbb{R} \Rightarrow x_n^2 \geq 0$$

$$\sum_n x_n^2 = 0 \Rightarrow x_n = 0 \quad \forall n$$

Consequence 2: Internal Generation of Sign Reversal

Classification: Derived consequence

$$x_1 = A, \qquad x_2 = iA$$

$$x_1^2 + x_2^2 = A^2 + (iA)^2 = A^2 - A^2 = 0$$

$$i^2 = -1$$

Consequence 3: Necessity of Phase Algebra

Classification: Derived consequence

$$x_n = r_n e^{i\phi_n}$$

$$x_n^2 = r_n^2 e^{i2\phi_n}$$

Closure condition:

$$\sum_{n=1}^{N} r_n^2 e^{i2\phi_n} = 0$$

Equal-amplitude condition:

$$r_n = r$$

$$\sum_{n=1}^{N} e^{i2\phi_n} = 0$$

Consequence 4: Type Anonymity

Classification: Derived consequence

$$x_n \in \mathbb{C} \quad \forall n$$

or

`all components are assigned the same phase-algebra type`

Consequence 5: Ninety-Degree Phase Difference

Classification: Derived consequence

$$A^2 + (iA)^2 = 0$$

---

3. Anonymous Equal-Amplitude Composite Wave

Axiom 4: Anonymous Equal-Amplitude Composite Wave

Classification: Axiom

$$\Psi_N = \frac{A_0}{N} \sum_{k=1}^{N} e^{i\theta_k}$$

$$|\psi_k| = \frac{A_0}{N}$$

When all phases are aligned:

$$\max |\Psi_N| = A_0$$

Axiom 5: Unreadability of Individual Phase

Classification: Axiom

$$\theta_k \to \theta_k + \alpha$$

Axiom 6: Phase-Difference Readout

Classification: Axiom

$$\Delta_{ij} = \theta_i - \theta_j$$

Number of readable pair channels:

$$\binom{N}{2} = \frac{N(N-1)}{2}$$

Axiom 7: System Identification

Classification: Axiom

$$(N, A_0, \{\Delta_{ij}\})$$

Axiom 8: Composite Phase Projection

Classification: Axiom

$$S_N = \sum_k e^{i\theta_k}$$

$$\Psi_N = \frac{A_0}{N} S_N$$

$$\Theta_N = \arg \Psi_N$$

Axiom 9: No Requirement of Inversion Symmetry

Classification: Axiom

Inversion symmetry is not required.

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4. External Readout

Axiom 10: External-Axis Response

Classification: Axiom candidate

$$M_\alpha = e^{i\alpha}$$

$$Y(\alpha) = \frac{A_0}{N} \sum_{k=1}^{N} e^{i(\theta_k - \alpha)} = e^{-i\alpha} \Psi_N$$

Continuous response:

$$S_N(\alpha) = \mathrm{Re} \left[\frac{1}{N} \sum_{k=1}^{N} e^{i(\theta_k - \alpha)} \right]$$

Binary response:

$$S_N(\alpha) = \mathrm{sgn}\, \mathrm{Re} \left[\frac{1}{N} \sum_{k=1}^{N} e^{i(\theta_k - \alpha)} \right]$$

Axiom 11: External Readout Phase

Classification: Axiom candidate

$$\Delta\phi_\mu = \phi_\mu^{(\Psi)} - \phi_\mu^{(M)}$$

$$\mu = x, y, z, t$$

Axiom 12: Position-Phase Readout

Classification: Axiom candidate

$$x, y, z$$

Axiom 13: Time-Phase Readout

Classification: Axiom candidate

$$\phi_t \equiv \omega t_{\text{read}} \pmod{2\pi}$$

Axiom 14: Covering Phase

Classification: Axiom candidate

```
2pi-periodic readout
4pi-periodic readout
winding-number-difference readout
```

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5. Observational Phase Discreteness

Axiom 15: Observational Phase Discreteness

Classification: Axiom candidate

$$\sum_{\text{cycle}} \Delta\phi = 2\pi m, \qquad m \in \mathbb{Z}$$

$$W = \frac{1}{2\pi} \oint d\phi, \qquad W \in \mathbb{Z}$$

Readout-cell form:

$$\phi_{\text{obs}} = m\epsilon_\phi, \qquad m \in \mathbb{Z}$$

Phase-area candidate:

$$Q_\phi = \sum_{i<j} \sin^2 \frac{\Delta_{ij}}{2}$$

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6. Waveform and Localization

Working Axiom A: State Waveform

Classification: Working hypothesis

$$A(\theta) = \sum_n A_n \cos(n\theta + \phi_n)$$

$$Z(\theta) = \sum_n r_n e^{i(n\theta+\phi_n)}$$

Definition A1: Normalized Equal-Amplitude Odd-Harmonic Localized Wave

Classification: Definition

$$N_h = 2K - 1$$

$$S_{N_h}(u) = \frac{1}{K} \sum_{m=0}^{K-1} \cos((2m+1)u)$$

$$Z_{N_h}(u) = \frac{1}{K} \sum_{m=0}^{K-1} e^{i(2m+1)u}$$

$$\Psi_{N_h}(u) = AS_{N_h}(u)$$

$$\Psi_{N_h}(u) = AZ_{N_h}(u)$$

$$\frac{A}{K}$$

Theorem A1: Representative-Amplitude Preservation

Classification: Derived theorem

$$S_{N_h}(0) = 1$$

$$\Psi_{N_h}(0) = A$$

Theorem A2: Highest Odd-Harmonic Order and Localization Width

Classification: Derived theorem

$$K = \frac{N_h + 1}{2}$$

$$\lambda_{N_h} = \frac{\lambda_0}{N_h}$$

$$\Delta x_{N_h} \propto \frac{\lambda_0}{N_h}$$

$$N_{h,\mathrm{req}} \sim \frac{\lambda_0}{\Delta x}$$

$$N_{h,\mathrm{req}}^{(t)} \sim \frac{T_0}{\Delta t}$$

Consequence A1: Identical Construction of Spatial and Temporal Localization

Classification: Derived consequence

$$\Psi(\chi,\tau) = AS_{N_{h,\chi}}(\chi-\chi_0)S_{N_{h,\tau}}(\tau-\tau_0)e^{i\phi}$$

Working Axiom B: Localization

Classification: Working hypothesis

$$S_{N_h}(u) = \frac{1}{K}\sum_{m=0}^{K-1}\cos((2m+1)u)$$

$$Z_{N_h}(u) = \frac{1}{K}\sum_{m=0}^{K-1} e^{i(2m+1)u}$$

Working Axiom C: Coupling Entrance

Classification: Working hypothesis

The entrance to interaction is low-order base-phase alignment.

Working Axiom D: Observation Map

Classification: Working hypothesis

$$s_\alpha = \mathrm{Loc}\left[\int A(\theta)M_\alpha(\theta)d\theta\right]$$

Working Axiom E: Squared Correlation Strength

Classification: Working hypothesis

$$P(\alpha) = \frac{|C_\alpha|^2}{\sum_\beta |C_\beta|^2}$$

Working Axiom F: Shared Low-Order Base Synchronization

Classification: Working hypothesis

$$\theta_A - \theta_B = \Delta$$

Working Axiom G: Synchronized Demodulation Failure

Classification: Working hypothesis

$$V(t) = \left| \sum_n w_n e^{i\epsilon_n(t)} \right|$$

$$V(t) = \left| \sum_n w_n e^{-\frac{1}{2}\sigma_n^2(t)} \right|$$

Working Axiom H: Hierarchy

Classification: Working hypothesis

```
low-order base mode
+ high-order localized mode
```

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7. Readout, Memory, and Complex Representation

I/Q Readout

Classification: Theoretical observation

$$z = a + ib$$

$$a = r\cos\theta, \qquad b = r\sin\theta$$

$$u = r\cos\theta + r\sin\theta = \sqrt{2}r\cos\left(\theta - \frac{\pi}{4}\right)$$

Harmonic-Information Readout

Classification: Theoretical observation

$$Z(\theta) = \sum_n r_n e^{i(n\theta+\phi_n)}$$

$$P(Z) = \mathrm{Re}(Z) + \mathrm{Im}(Z)$$

$$P(Z) = \sum_n \sqrt{2}r_n \cos\left(n\theta + \phi_n - \frac{\pi}{4}\right)$$

Preservation Side and Readout Side

Classification: Theoretical observation

$$z\bar{z} = a^2 + b^2$$

$$z^2 = a^2 - b^2 + 2iab$$

Square Registry

Classification: Working hypothesis

$$\mathcal{A}_0 = \sum_k A_k^2$$

$$a_0 = \sum_k |A_k|^2 = \sum_k A_k \bar{A}_k$$

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8. Statistical Classification

Pair Phase Difference

Classification: Definition

$$\Delta_{ij} = \theta_i - \theta_j$$

$$\binom{N}{2} = \frac{N(N-1)}{2}$$

Pair composite:

$$\psi_i + \psi_j = 2A \cos \frac{\Delta_{ij}}{2} e^{i(\theta_i + \theta_j)/2}$$

Pair interference strength:

$$I_{ij} = 4A^2 \cos^2 \frac{\Delta_{ij}}{2}$$

Overlap Degree

Classification: Definition

$$R_N = \left| \frac{A_0}{N} \sum_{k=1}^{N} e^{i\theta_k} \right|$$

$$\rho_N = \frac{R_N}{A_0} = \frac{1}{N} \left| \sum_{k=1}^{N} e^{i\theta_k} \right|$$

$$0 \le \rho_N \le 1$$

Fermionic Degree

Classification: Definition

$$F_N = \frac{2}{N(N-1)} \sum_{i<j} \sin^2 \frac{\Delta_{ij}}{2}$$

$$B_N = 1 - F_N$$

$$B_N = \frac{2}{N(N-1)} \sum_{i<j} \cos^2 \frac{\Delta_{ij}}{2}$$

$$F_N = \frac{N}{2(N-1)}(1 - \rho_N^2)$$

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9. Observation Selection and Curvature Projection

Axiom 16: Unique Observation Selection by Complete Pairwise Relational Waves

Classification: observation-connection axiom

This axiom is an independent observation principle not derived from Axiom 0 or Axiom 1.

For a closed system with component count `N`, define the unordered complete pairwise relation set without proper names as

$$\mathcal{E}_N = \{\{i,j\} \mid 1 \le i < j \le N\}$$

For each relation

$$e = \{i,j\} \in \mathcal{E}_N$$

the corresponding

$$X_e \in \mathbb{C}$$

is a physical relational wave that constitutes the system, not an observation-device channel.

The relational wave `X_e` is not a quantity derived from component waves; it is an independent physical state component. The correspondence map to component waves is not specified by this axiom.

Write the relational-wave state as

$$X_{\mathcal{E}} = (X_e)_{e \in \mathcal{E}_N}$$

A permutation of individual names acts as a permutation of relational waves and does not change the closure equation, the update rule, or the observation selection.

Define the quadratic form of the relational-wave set as

$$q_{\mathcal{E}}(X) = \sum_{e \in \mathcal{E}_N} X_e^2$$

The closure of the relational waves is written, using the compensation quantity of Axiom 3, as

$$\sum_{e\in\mathcal{E}_N} X_e^2 + (iR)^2 = 0$$

The transposed form

$$q_{\mathcal{E}}(X) = R^2$$

may be used.

Let the number of relational waves be

$$M = |\mathcal{E}_N|$$

and let a real-coefficient generator acting identically on all relational waves be

$$K \in \mathbb{R}^{M\times M}$$

The relational wave `X_e` remains complex, and `K` acts as the same real-coefficient transformation on its real and imaginary parts.

Write the local linear update of relational waves as

$$\dot{X} = KX$$

If this update preserves the quadratic form identically,

$$\frac{d}{d\tau} q_{\mathcal{E}}(X) = X^{\mathsf{T}}\left(K^{\mathsf{T}} + K\right)X = 0$$

Therefore, the closure-preserving generator of relational waves satisfies

$$K^{\mathsf{T}} = -K$$

This antisymmetry is not introduced from any particular physical name or axis name.

A real antisymmetric generator decomposes, without additional proper names, privileged axes, or external references, into a direct sum of two-dimensional rotation planes and a null space.

Let the number of rotation planes be

$$r = \frac{1}{2}\operatorname{rank} K$$

and the null-space dimension be

$$\dim\ker K = M - 2r$$

The adoption conditions for intrinsically observable directions are as follows.

First, rotation planes become candidates for distinct directions only when they can be mutually distinguished by the relational quantity of independent rotation frequencies. Distinguishability is a necessary condition and does not by itself license adoption as spatial directions.

Second, when a normal direction is read from the plane spanned by two linearly independent relational directions, the normal may be adopted as a direction unique up to sign only when the normal-candidate space is one-dimensional. For a real `d`-dimensional spatial display, the dimension of the normal-candidate space is

$$d - 2$$

and this condition holds only for

$$d = 3$$

Third, the number of intrinsically observable directions is not counted from the relational-wave count `M`, the generator rank, or the null-space dimension. It is counted as the number of directions satisfying the uniqueness conditions above.

The minimal example is `M=3`: when

$$\operatorname{rank} K = 2, \qquad \dim \ker K = 1$$

the two directions of one rotation plane and the one normal direction uniquely determined from that plane, three directions in total, may be adopted.

When `M=6` and three rotation planes are distinguished by independent rotation frequencies, three directions may be adopted.

When the number of distinguishable rotation planes exceeds the range of spatial display satisfying the second uniqueness condition, the excess rotation planes are not adopted as spatial directions and are retained as internal phase modes.

When multiple isomorphic rotation planes remain, or when a normal-candidate space or null space of dimension two or more remains and its internal directions cannot be distinguished by relational quantities alone, no single direction among them may be selected as an observable spatial direction. Such a residue is retained as an internal subspace without a unique basis.

The primitive readout of each relational wave is constructed from the phase difference of Axiom 6,

$$\Delta_{ij} = \theta_i - \theta_j$$

and the bilinear relation obtained from the quadratic form of Axiom 1,

$$q(X) = \sum_n X_n^2$$

$$B(U, V) = \frac{1}{2}\left(q(U + V) - q(U) - q(V)\right)$$

Readout for three or more bodies is constructed from the set of complete pairwise relational waves and the invariant decomposition of their closure-preserving generator.

No independent higher-order readout rule may be added on the grounds of physical names, object names, or dimension names.

The specific coupling strengths, phase-difference dependence, and update period of the relational-wave generator are not specified by this axiom.

They are defined as independent relational update rules consistent with Axiom 0, Axiom 0+, Axiom 1, and this axiom.

Working Axiom I: Phase-Difference Sine Relational Update Rule

Classification: working hypothesis

As a concrete form of the relational-wave generator of Axiom 16, define the endpoint-sharing adjacency

$$A_{ef} = \begin{cases} 1, & e \neq f \text{ and } e \cap f \neq \varnothing, \\ 0, & \text{otherwise} \end{cases}$$

and, from the initial phases `theta_e = arg X_e`, set

$$\widetilde{K}_{ef} = A_{ef} \sin(\theta_f - \theta_e)$$

It satisfies

$$\widetilde{K}^{\mathsf{T}} = -\widetilde{K}$$

and thus the antisymmetry condition of Axiom 16.

This coupling rule is not derived from Axiom 0, Axiom 0+, Axiom 1, or Axiom 16. It is a working hypothesis placed as one concrete closure-preserving relational update rule.

Consequence 16.1: Vertex Decomposition and Linear Rank Bound

Classification: derived consequence (under Working Axiom I)

Let `B` be the unsigned incidence matrix with individuals as rows and relations as columns, and from its `k`-th row build

$$c_k = (B_{ke} \cos \theta_e)_e, \qquad s_k = (B_{ke} \sin \theta_e)_e$$

Then

$$\widetilde{K} = \sum_{k=1}^{N} \left(c_k s_k^{\mathsf{T}} - s_k c_k^{\mathsf{T}} \right)$$

holds exactly.

Each term is an antisymmetric matrix of rank at most 2, so for all `N` and all initial phases,

$$\operatorname{rank} \widetilde{K} \leq 2 \min\left(N, \left\lfloor \frac{M}{2} \right\rfloor \right)$$

The number of relational waves

$$M = \binom{N}{2}$$

grows quadratically, while the number of rotation modes

$$r = \frac{1}{2} \operatorname{rank} \widetilde{K}$$

grows at most linearly.

Proof: Paper 3 of the Dimensional Generation Structure series. Concept DOI: 10.5281/zenodo.21465898

Consequence 16.2: Generic-Position Rank Equality

Classification: derived consequence (computer-assisted proof, `3 <= N <= 12`)

For each `N` with `3 <= N <= 12`, for all initial phases except a set of Lebesgue measure zero,

$$\operatorname{rank}\widetilde{K} = 2\min\left(N, \left\lfloor \frac{M}{2} \right\rfloor\right)$$

The proof combines exact rational witness configurations via the tan half-angle parametrization with the fact that the zero set of a nontrivial real-analytic function has Lebesgue measure zero. Paper 3. Concept DOI: 10.5281/zenodo.21465898

The generic null-space dimension is then

$$\dim\ker\widetilde{K} = M - 2\min\left(N, \left\lfloor \frac{M}{2} \right\rfloor\right)$$

which for `N >= 5` equals

$$\frac{N(N-5)}{2}$$

Equality for all `N` is a new hypothesis and is not included in this consequence.

Consequence 16.3: Normal Uniqueness

Classification: derived consequence

When a normal direction is reconstructed from the plane spanned by two linearly independent relational directions, without additional background axes or selection rules, the normal-candidate space of a real `d`-dimensional display has dimension

$$d - 2$$

and the normal is unique up to sign only for

$$d = 3$$

For `d >= 4`, the orthogonal group

$$O(d-2)$$

acting on the normal-candidate space while fixing the plane leaves a continuous selection freedom, and under anonymity (Axiom 0) no single direction can be selected. The projector and squared quantities are unique, but the individual linear directions inside are not.

This consequence is the basis of the second adoption condition of Axiom 16. Proof: Paper 3. Concept DOI: 10.5281/zenodo.21465898

Axiom 17: Curvature Reaction Projection Centered on the Future Phase Position

Classification: dynamical projection axiom

This axiom is an independent dynamical and projection principle not derived from Axiom 0 or Axiom 1.

Assign an order index

$$\tau \in \mathbb{N}$$

to the state sequence of a closed system.

In the state update

$$X(\tau) \longrightarrow X(\tau + 1)$$

the phase position on the `tau+1` side is defined as the future phase position of that state sequence.

When the phase progression on a rotation plane uniquely constructed without additional names by Axiom 16 has a curvature-radius readout `rho_c` and an angular-velocity readout `omega`, and can be expressed as a motion whose relational virtual rotation center is the future phase position, the center-directed compensation that closes this virtual rotation — that is, the reaction to the centrifugal force — is mapped and read out as an observable acceleration in the circumferential direction of progression.

The center-directed compensation is the internal representation, and the circumferential acceleration is the external readout. This axiom defines the projection connecting the two as a rule.

The magnitude of the acceleration is

$$|\alpha_{\text{read}}| = \rho_c |\omega|^2$$

Let `t_hat_tau` be the unit vector of the circumferential direction of progression pointing from the current phase position toward the future phase position. The readout acceleration is

$$\boldsymbol{\alpha}_{\text{read}} = \rho_c |\omega|^2 \, \hat{\boldsymbol{t}}_\tau$$

Here `rho_c`, `omega`, and `t_hat_tau` are relational readout quantities without physical names in the first-principles layer. The circumferential direction is not an absolute axis of a background space; it is determined each time from the relation among the current phase position, the future phase position, and the closure radius.

This axiom may not be modified on the grounds of gravity, Coulomb force, mass, charge, or any other physical name.

The same projection rule applies to every closure mode satisfying the same conditions.

If the phase branch opposite to the future side, a reversal of the acceleration direction, or sign retention or sign elimination is adopted, it must not be selected from physical names; it must be defined in advance as an independent phase relation, branch selection, symmetry breaking, or readout operation.

This axiom does not identify the readout acceleration with standard gravity, the standard Coulomb force, or any known external force.

Physical names are given only after the acceleration map, distance exponent, sign, closure preservation, and gauge stability have been confirmed.